Distance and Buffers

Lecture #13 | GEOG 510
GIS & Spatial Analysis in Public Health

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Outline

- Distance
  - Euclidean distance
    - Distance to features
    - Buffer
  - Alternate distances
    - Great circle, Manhattan, Network, Cost distance (raster)

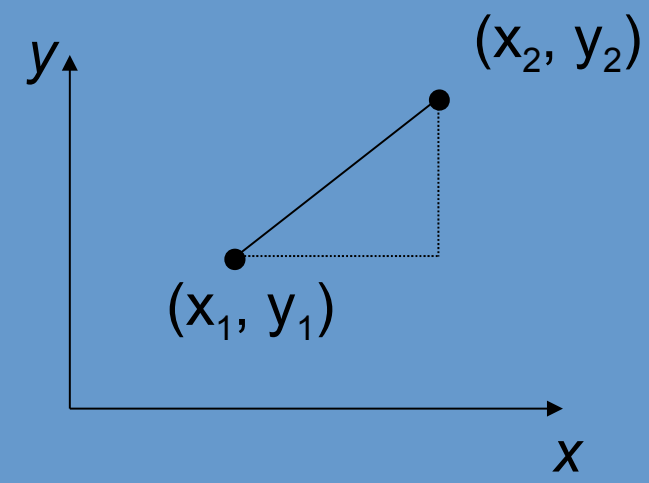
Distance

- Why distance?
  - Distance is a *fundamental geographic concept* that affects relationships and interactions among phenomena
    - Sometime we want to explain it
    - Sometimes we want to measure it for input to other calculations
    - Sometimes we want to map it

Euclidean Distance

- Based on Pythagorean Theorem
  - $a^2 + b^2 = c^2$
  - Assumes:
    - A planar Earth (i.e., a projected cs)
    - No differences in elevation

Does not require GIS!... simple mathematical calculation

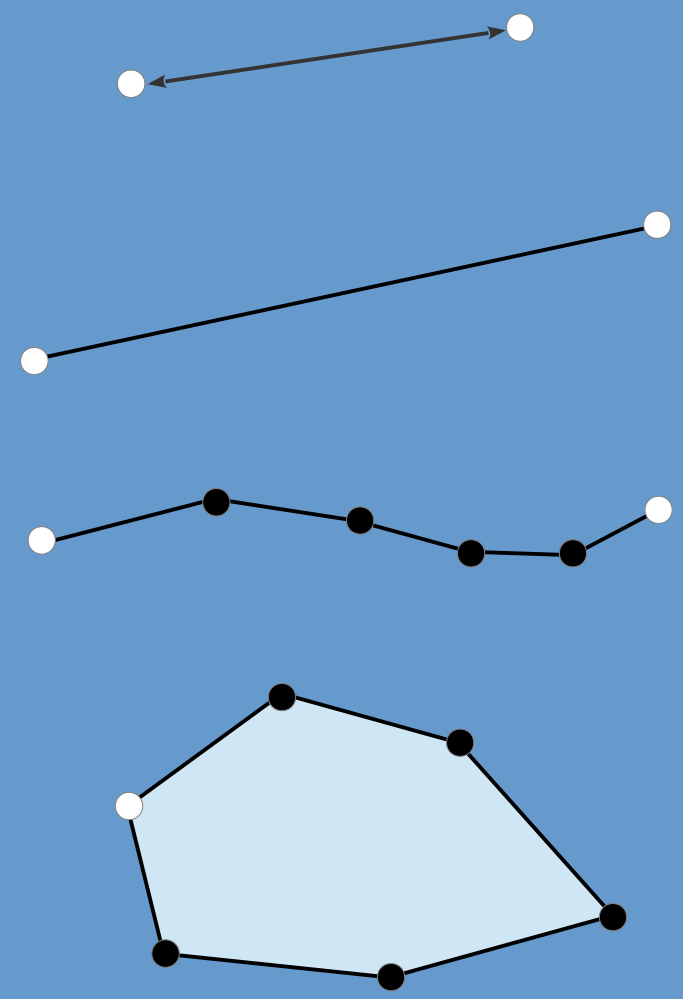


$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Euclidean distance is also called "Cartesian" distance in ESRI help

Distance: Vector

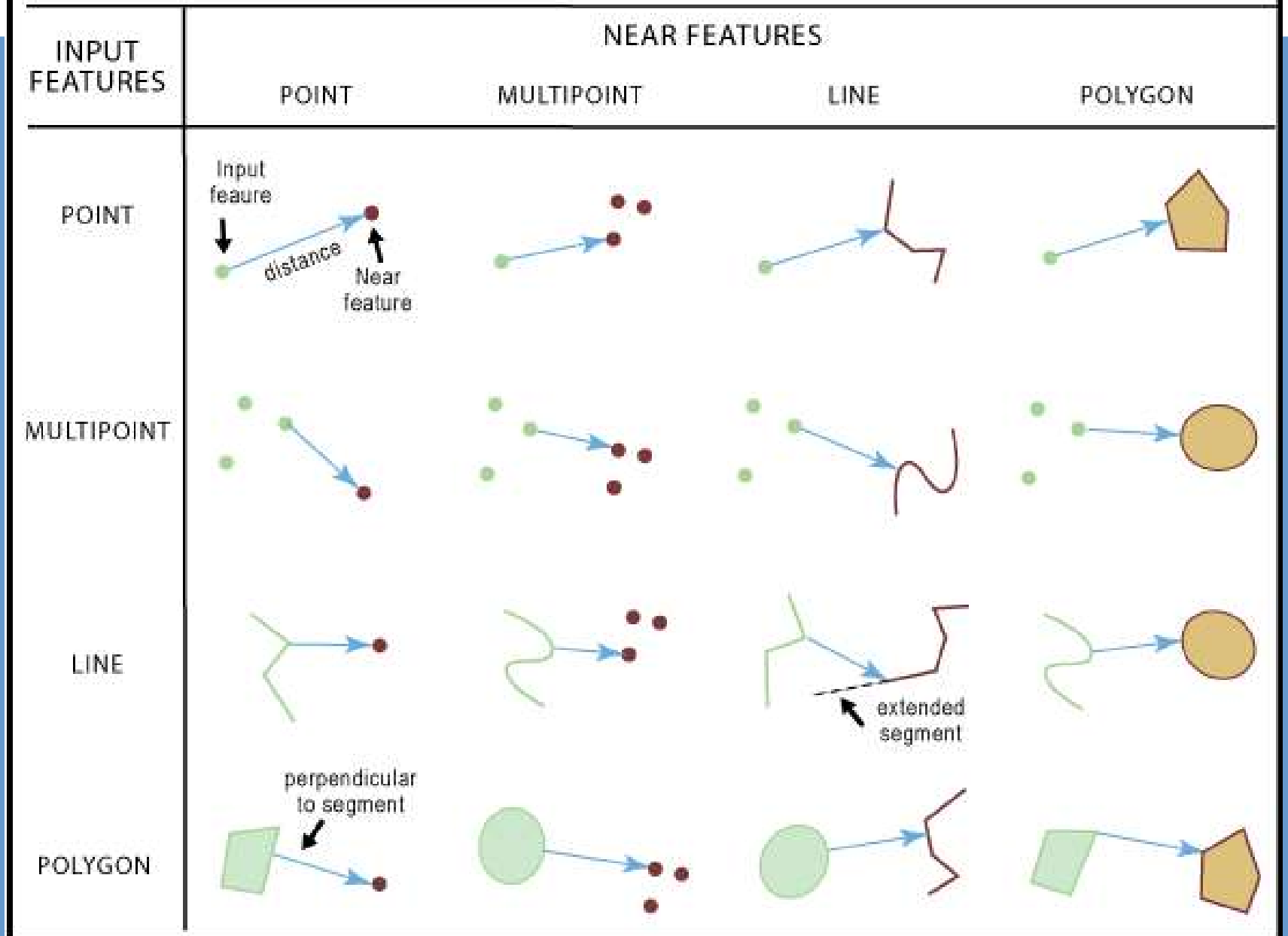
- Point distance
  - Distance between points
- Line length
  - Distance between nodes
  - Distance between vertices
    - Sum of segment lengths
- Polygon Perimeter
  - Distance between vertices
    - Sum of segment lengths



Distance to Features

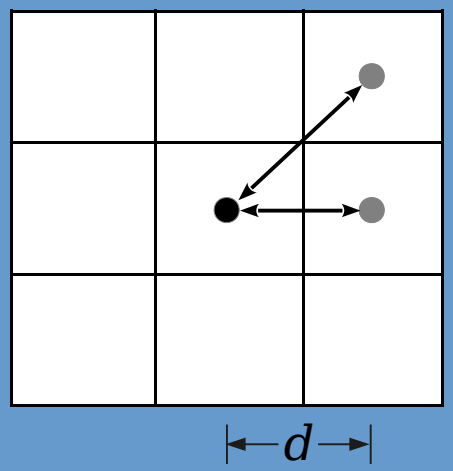
- Measure distance from one set of features to another
  - Vector distance tool
  - Near
    - Locate nearest feature, return distance
  - Generate Near Table
    - Locate *k* nearest features, return distance
      - Output written to a new table

HOW DISTANCES ARE CALCULATED BETWEEN FEATURES



Distance: Raster

- Measured from some source cell
  - Or, source cells (could be a line or polygon)
  - Distance from each cell to source calculated as Euclidean distance

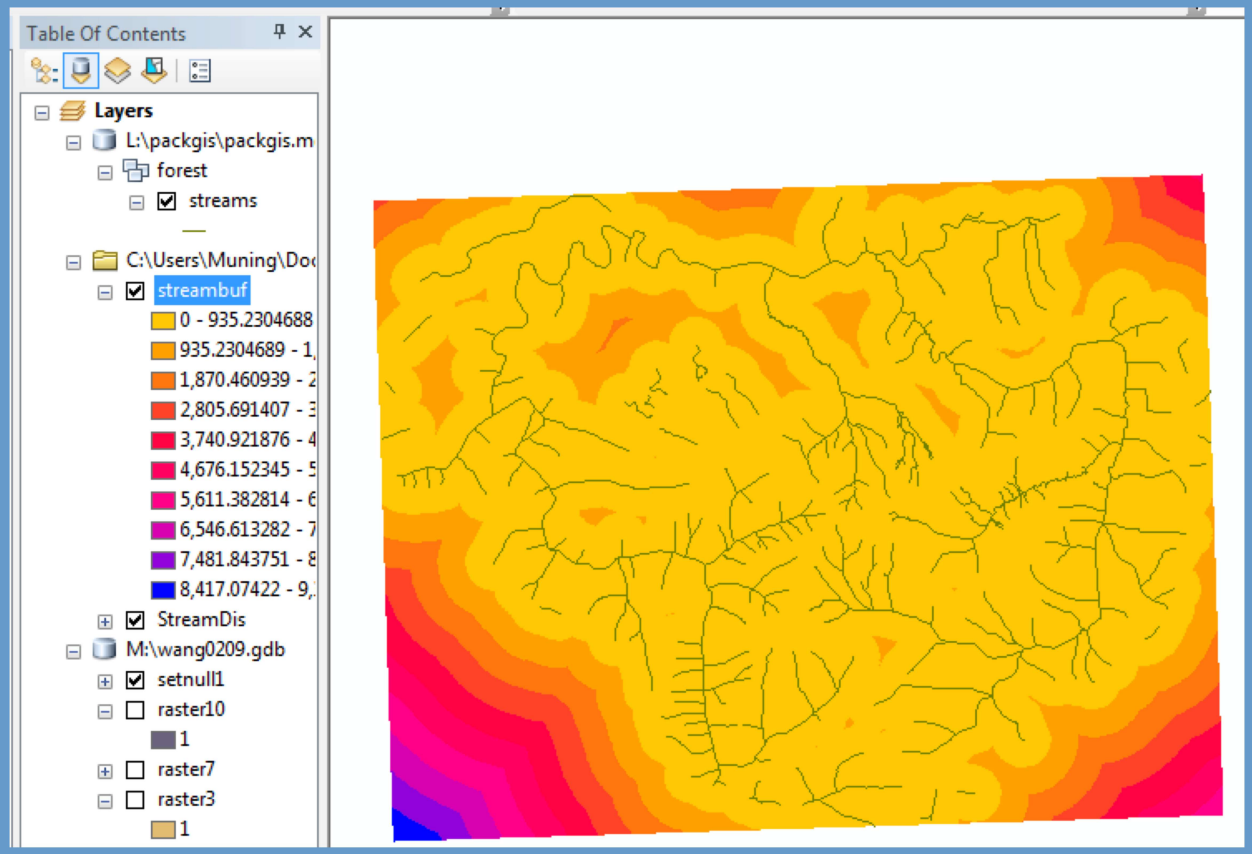


1.4	1	1.4
1	0	1
1.4	1	1.4

2.8	2.2	2	2.2	2.8
2.2	1.4	1	1.4	2.2
2	1	0	1	2
2.2	1.4	1	1.4	2.2
2.8	2.2	2	2.2	2.8

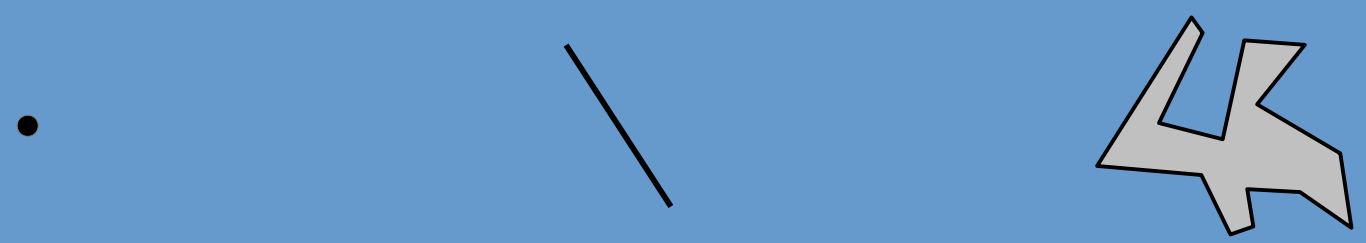
Distance to Features

- Raster maps of distance are commonly used as measures of accessibility



Buffers

- A buffer is a region surrounding a feature (or set of features)
  - Region (or neighborhood) defined using a set distance from a given feature (point, line, polygon, grid cell, or group of cells)
  - Create *new spatial objects* defined as the area within a *specified distance* from existing features



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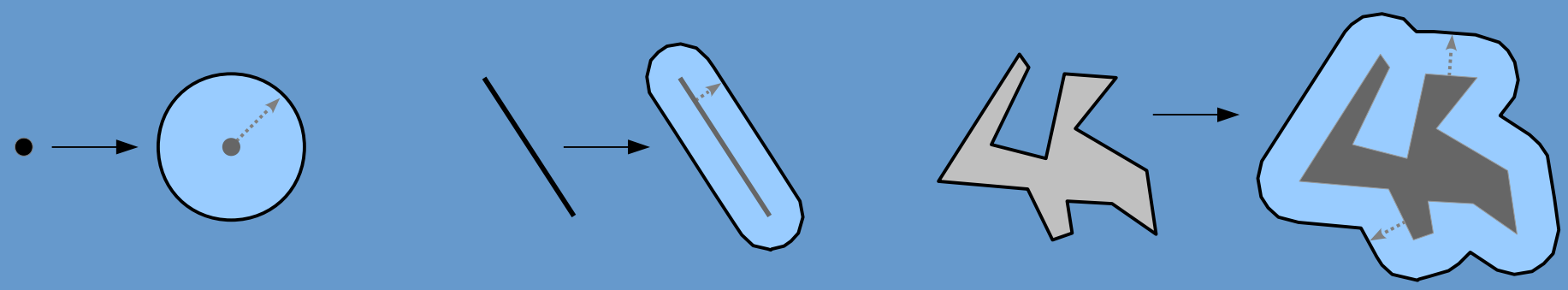

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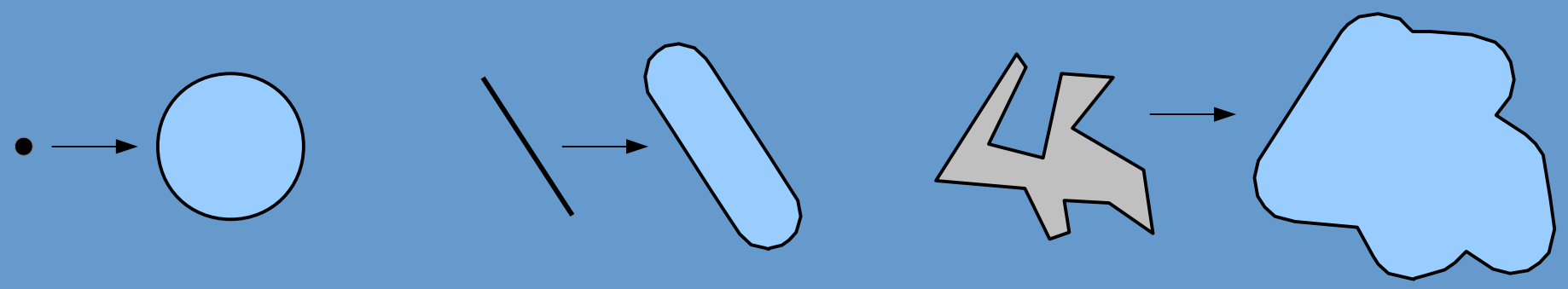
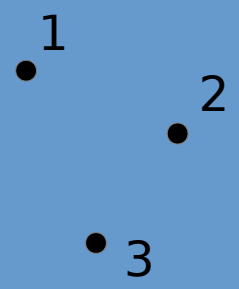
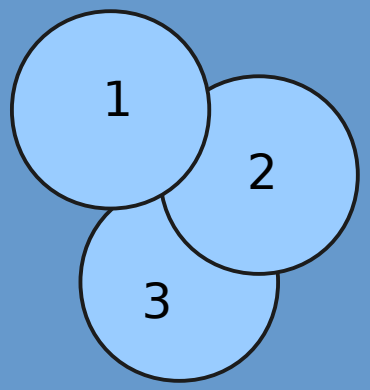


Buffer Output

- Output shape options for overlapping buffers in same feature class
  - An *autonomous* buffer for each feature
    - Retains input features' attributes

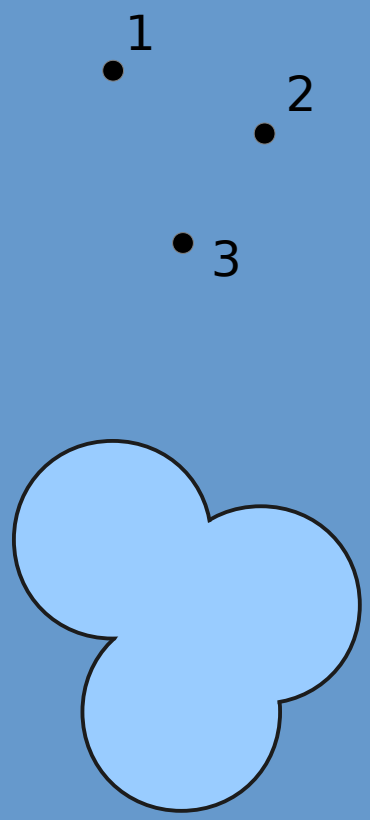




Planar topology?

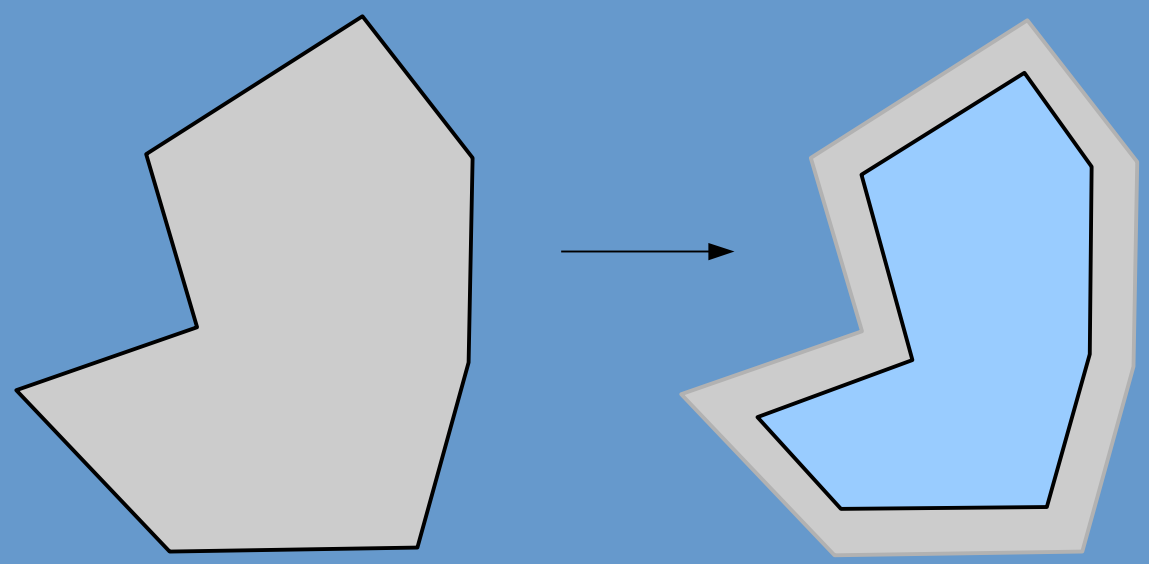
Buffer Output

- Output shape options for overlapping buffers in same feature class
  - An *autonomous* buffer for each feature
    - Retains input features' attributes
  - Dissolved output
    - Original features' attributes cannot be "separated"



Interior Buffer

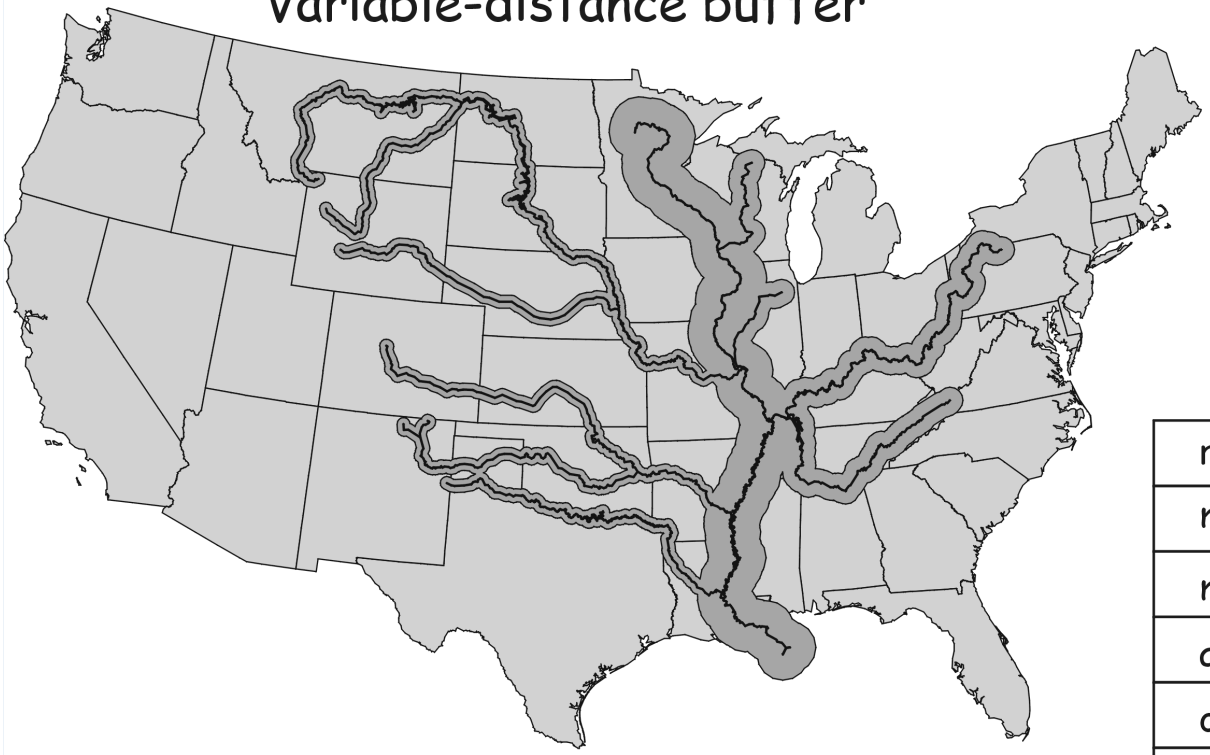
- Interior polygon buffers
  - Rather than choose a distance "outside" of the polygon, an interior buffer ("setback") can be created
    - Use a "negative" value for the buffer distance



Variable Distance Buffer

- Buffer distance can be the same for all objects in a feature class (*fixed-distance*)
  - Buffer distance can also vary
    - Variable-distance buffer
    - Based on an *attribute* in the feature class

Variable-distance buffer



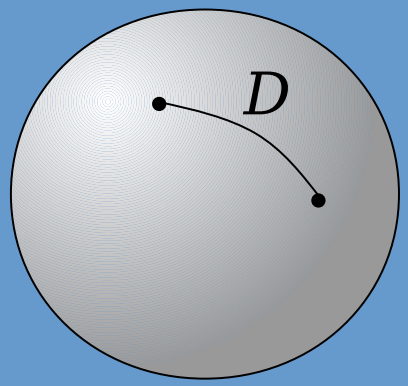
river_identifier	buffdist
mississippi	100
missouri	50
arkansas	50
ohio	75
tennessee	75
st. croix	75
illinois	75
wisconsin	75

Euclidean Distance

- Limitations
  - Assumes flat earth
    - Especially limiting over large distances
  - Assumes no slope in terrain
    - Especially limiting in mountainous terrain
  - Assumes no barriers to movement
    - Barriers can include anything that's not a road (in the case of vehicular travel) or anything on the landscape that stands in the way

Great Circle Distance

- Great circle distance is the shortest distance between two points on a sphere
  - Used for points stored in spherical coordinates
  - Required to calculate distance in spherical coordinates, especially if points are far apart



Point 1: $(\lambda_1, \Phi_1) \leftarrow$ longitude, latitude

Point 2: $(\lambda_2, \Phi_2) \leftarrow$ longitude, latitude

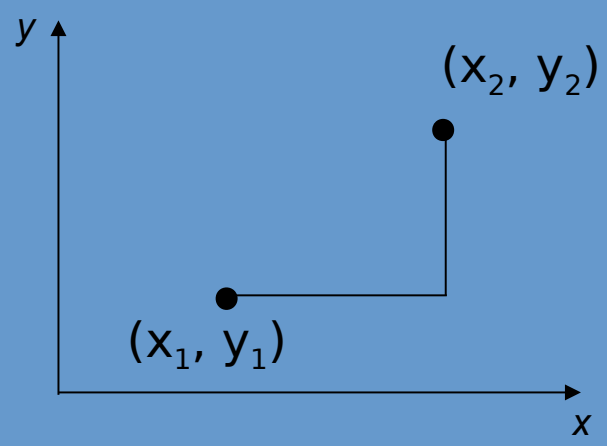
R: radius of the Earth (~6378km)

$$D = R \cdot \cos^{-1}((\sin\phi_1 \cdot \sin\phi_2) + (\cos\phi_1 \cdot \cos\phi_2 \cdot |\lambda_1 - \lambda_2|))$$

Also called "geodesic" distance in ESRI help

Manhattan Distance

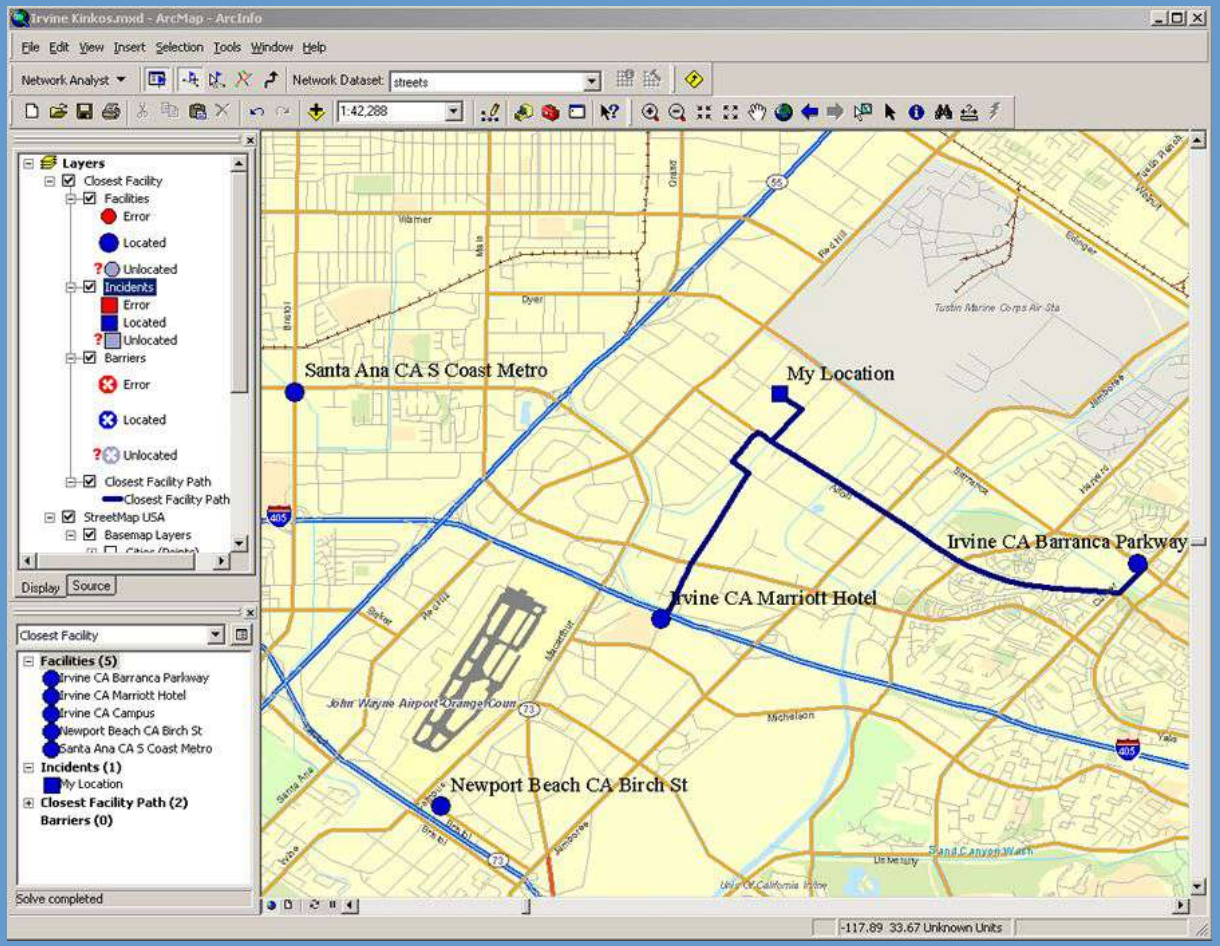
- One way to account for barriers is to recognize that travel can be limited to transportation routes that follow cardinal directions (N, S, E, W)
  - e.g., in Manhattan where most streets are at right angles



$$d = |x_2 - x_1| + |y_2 - y_2|$$

Network Distance

- Network distance restricts travel along existing arcs (lines)
  - e.g., on city streets
  - Distance is the sum of the lengths of arcs that make up the route



Minkowski Distance

- Falls between Euclidean and Manhattan distance
- Distance of a curved arc connecting two points

$$d=\sqrt{(x_2-x_1)^2+(y_2-y_2)^2}$$

$$d=|x_2-x_1|+|y_2-y_2|$$

$$d=\sqrt[p]{(x_2-x_1)^p+(y_2-y_2)^p}$$

where 1 < p < 2

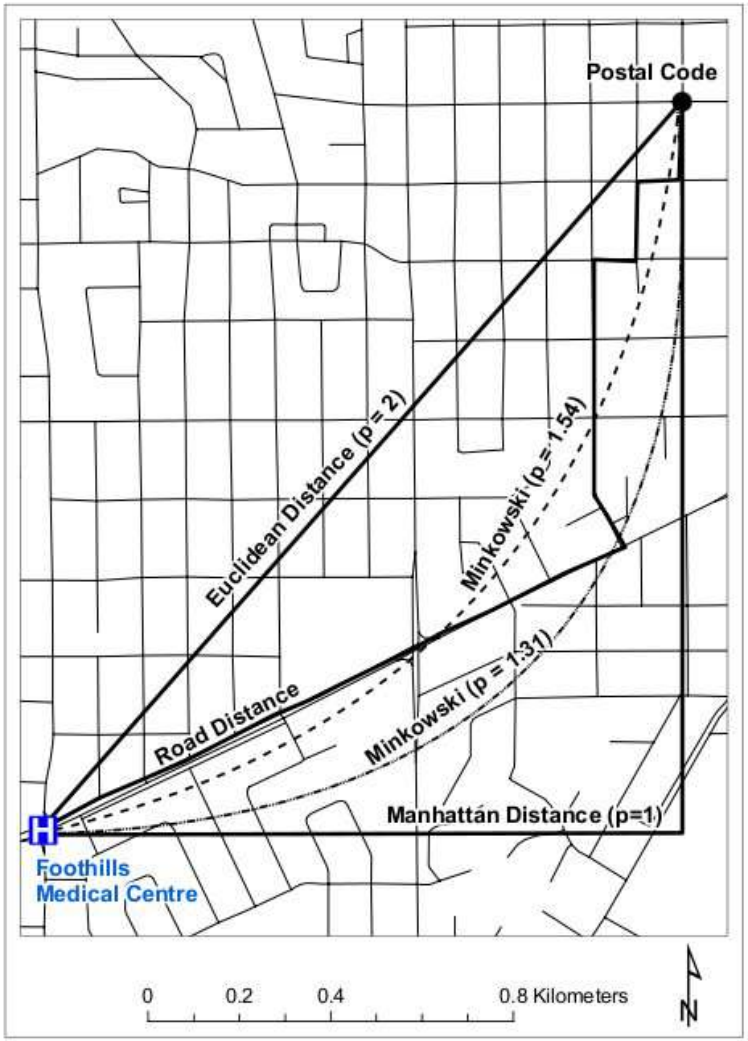
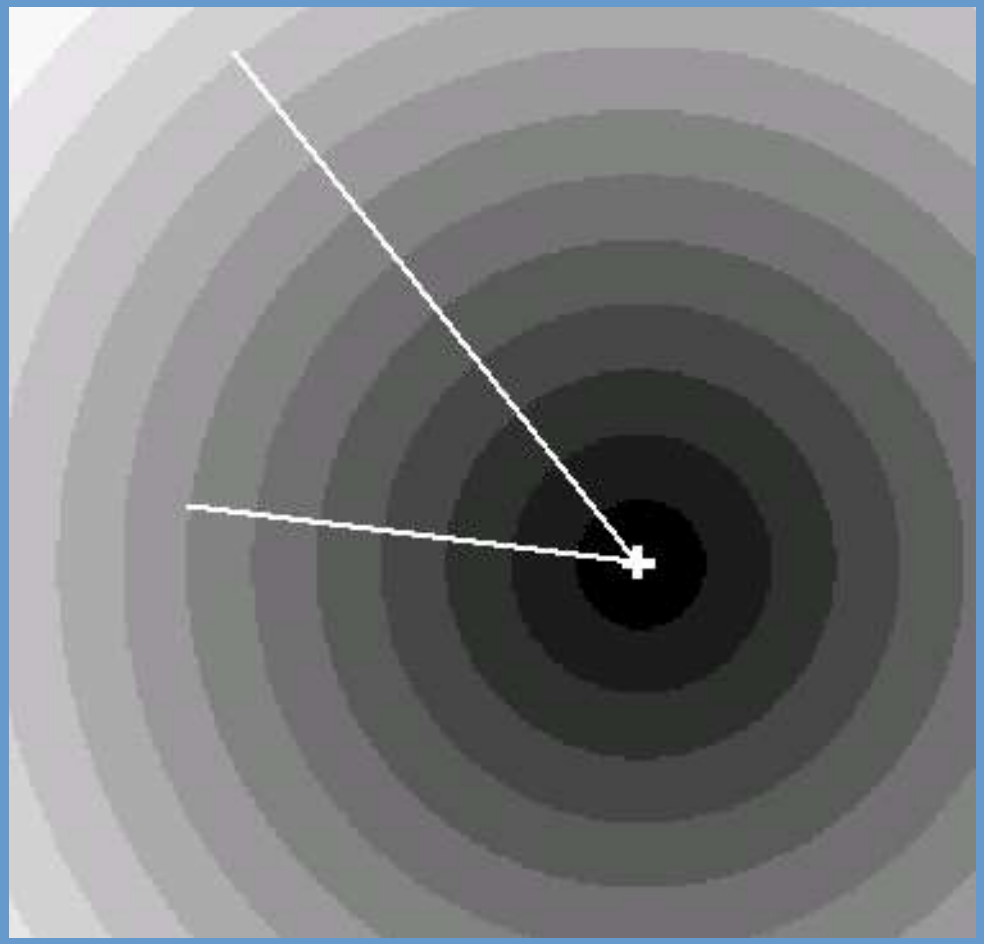


Figure 1
Visual illustration of road distance and distance metrics.

Shortest Path (Raster)

- Greater distances are in lighter shades
- Straight lines emanating from source location



Cost Distance

- Implemented as a modification to calculating raster distance
  - *Functional* or *cost* distance
    - User defines relative barriers to movement by mapping the cost, or friction, of movement for *each raster cell*
    - Cost can be defined by any *spatial feature* or *attribute* that limits the ability of people, animals, ideas, diseases, air, etc. to move across the landscape

Cost Distance

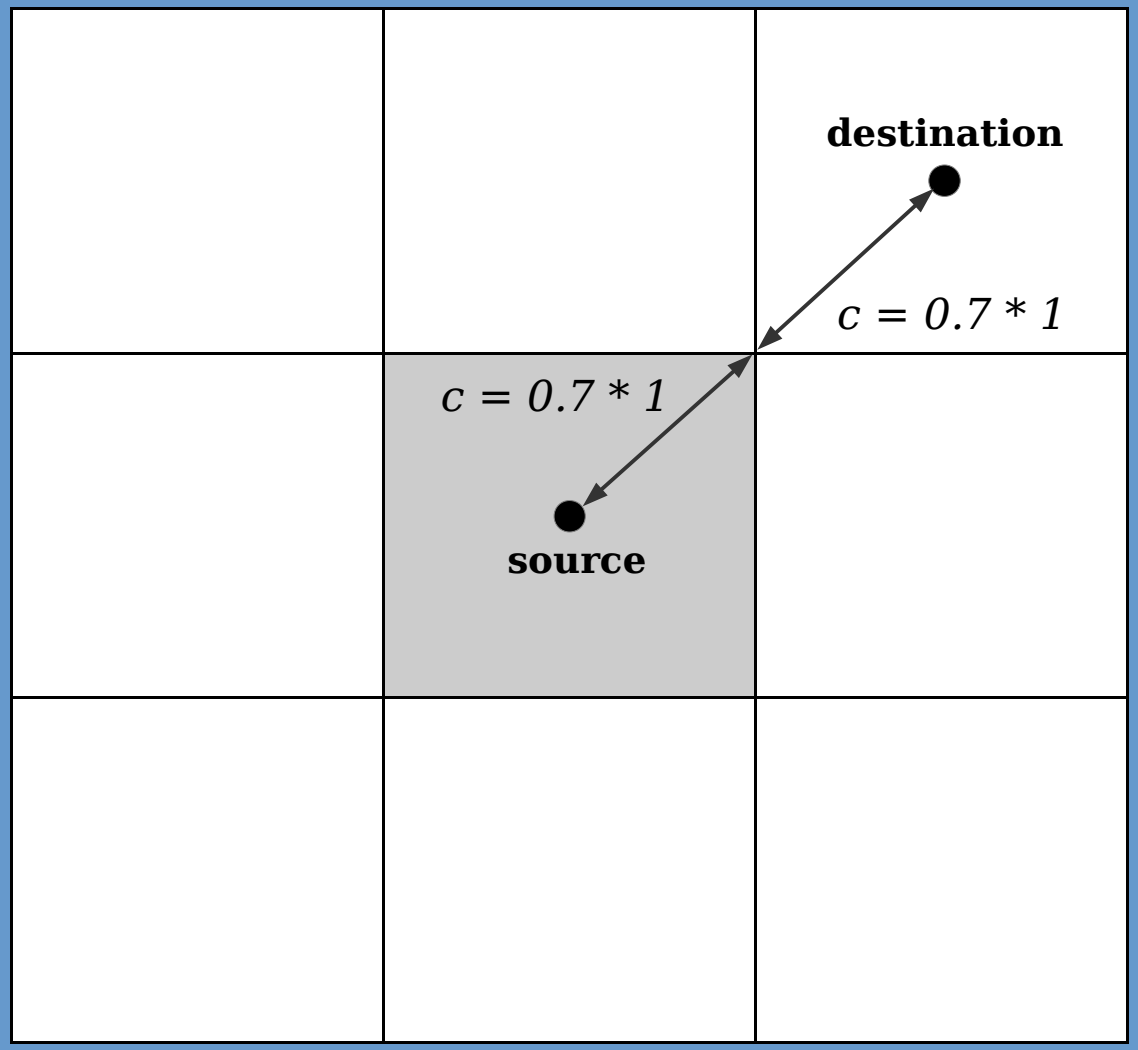
Cost

2	1	1
2	1	1
2	1	1

Distance

1.4	1	1.4
1	0	1
1.4	1	1.4

TOTAL COST, c = 1.4



Calculate Cost Distance

- Create a grid of *costs* (cost grid) to define the relative cost of traversing each grid cell
- Identify source cell(s)
- Calculate "cost distance" as the sum of unit costs accumulated to get from one place to other places
- Identify least cost path
  - i.e., the path of minimum cost between places

3.4	2.4	2	2.4	2.8
3.8	3.5	1	1.4	2.4
4.8	2.5	0	1	2
3.8	3.5	1	1.4	2.4
3.4	2.4	2	2.4	2.8

Cost distance (minimum to all cells in grid)

1	1	1	1	1
1	4	1	1	1
1	4	1	1	1
1	4	1	1	1
1	1	1	1	1

Cost grid

0	1	0	0	0
1	0	1	0	0
0	0	1	0	0
0	0	0	0	0
0	0	0	0	0

3.4	2.4	2	2.4	2.8
3.8	3.5	1	1.4	2.4
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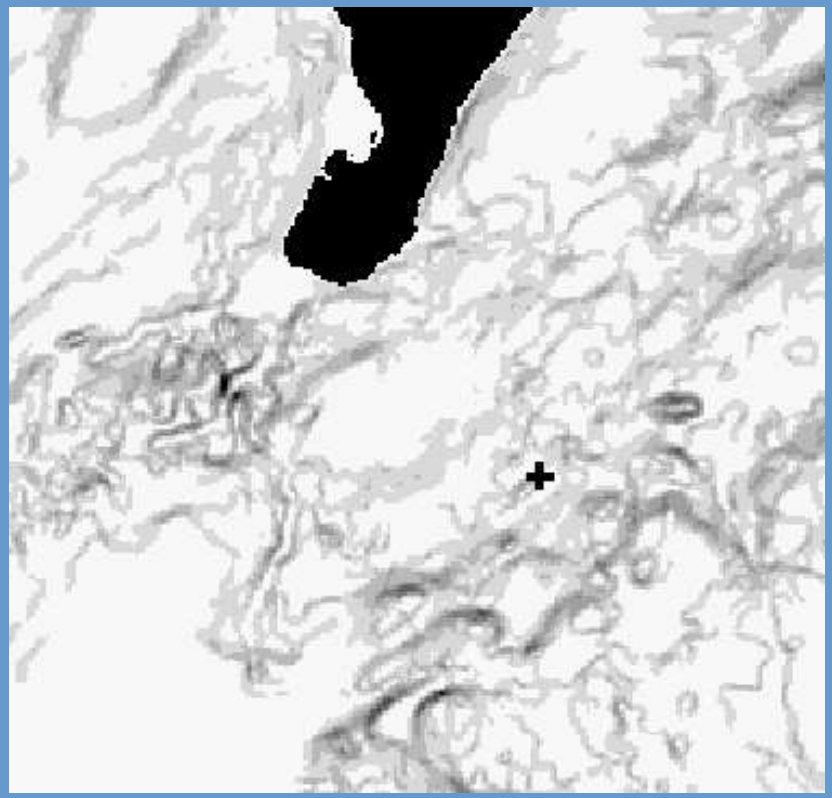
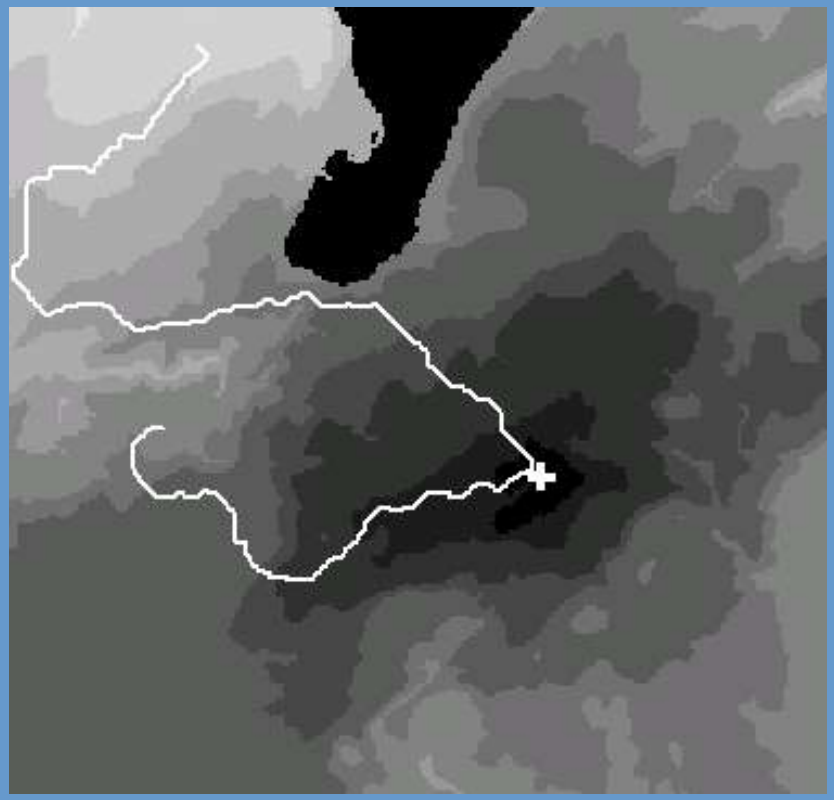
3.8	3.5	1	1.4	2.4
3.4	2.4	2	2.4	2.8

Least cost path

Cost Distance Example





- Cost grid: Slope
- Slope value is relative barrier (higher is darker)
- Lake is absolute barrier

- Cost Distance and Least Cost Path
- Greater accumulated cost distance is in lighter colors

Cost Distance

- Catchment area (like buffer)
  - Area within a specific "cost" of source
    - e.g., cost < 3.5

3.4	2.4	2	2.4	2.8
3.8	3.5	1	1.4	2.4
4.8	2.5	0	1	2
3.8	3.5	1	1.4	2.4
3.4	2.4	2	2.4	2.8

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3.8	3.5	1	1.4	2.4
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Keywords

- Buffer
- Distance
  - Euclidean, Great circle, Manhattan, Network, Minkownski
  - Least Cost Path
    - Functional or cost distance